

Water Treatment, E. coli Reduction, and Diarrhea: Causal Inference Using Double/Debiased Machine Learning

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December 26, 2025

Abstract

This paper investigates the 1) causal impact of household water treatment on E. coli contamination and 2) E.coli contamination of drinking water on child diarrhea using data from 59,633 households across 25 countries in Sub-Saharan Africa, Latin America, and Asia. While household water treatment is widely promoted as a preventive measure, estimating its true effect is challenging due to endogeneity stemming from household self-selection and the lack of objective water quality data in most household surveys. To address these issues, we apply Double/Debiased Machine Learning (DDML)—a flexible causal inference framework—to control for high-dimensional confounders and mitigate bias in the estimated treatment effects.

- Conceptualization (Akito - already semi done),
- Literature review (Akito - already semi done)
- Data: Downloaded, cleaned, and pooled the data (May need some update)
- Analysis (Need more work on specification and running analysis)
 - How to consider strain and settle
 - Water treatment of the water tested vs general
- Paper draft

Funding: No funding was received for this research.

JEL Classification: I15

Keywords: Water Treatment, Water Quality, WASH, E. Coli, Sub-Saharan Africa

[Dropbox](#)

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Journal

- High: The Lancet Global Health
- High: Journal of Human Resources
Bennett, D. (2008). Does clean water make you dirty? Water supply and sanitation in the rural Philippines.
- Journal of Health Economics (High and not good match?)
- Mid-high: Economic Development and Cultural Change
- Mid-high: World Development
- Mid-high: World Bank Research Observer
<https://academic.oup.com/wbro/article-abstract/22/1/1/1655339>
- Mid-high: World Bank Economic Review
- Mid-high: Oxford
- BMC Public Health
- PLOS ONE
- Oxford Development Studies
- Review of Economics of Household
- Environmental Health Perspectives – 10.1 (Robert Bain)
- BMC Public Health
- BMJ
Clasen, T., Schmidt, W.-P., Rabie, T., Roberts, I., & Cairncross, S. (2007). Interventions to improve water quality for preventing diarrhoea: systematic review and meta-analysis. BMJ, 334(7597), 782. <https://doi.org/10.1136/bmj.39118.489931.BE>.

LANCET Specific

0.1 Research in context

Evidence before this study This section should include a description of all the evidence that the authors considered before undertaking this study. Authors should briefly state: the sources (databases, journal or book reference lists, etc) searched; the criteria used to include or exclude studies (including the exact start and end dates of the search), which should not be limited to English language publications; the search terms used; the quality (risk of bias) of that evidence; and the pooled estimate derived from meta-analysis of the evidence, if appropriate.

0.2 Added value of this study

Authors should describe here how their findings add value to the existing evidence.

0.3 Implications of all the available evidence

Authors should state the implications for practice or policy and future research of their study combined with existing evidence. Research in context panels should not contain references; key studies mentioned here should be referenced in the main text.

1 Introduction

Each year, an estimated 842,000 people die from diarrheal diseases due to unsafe drinking water, poor sanitation, and inadequate hand hygiene. According to World Health Organization (WHO) estimates, 361,000 of those deaths are children under five, whose deaths could be prevented by addressing these risk factors. To combat these preventable deaths, governments from a lot of countries started increasing the availability of safe water sources by investing heavily in taps, tube wells, and hand pumps that reduce exposure to waterborne pathogens such as fecal coliform by accessing groundwater protected from surface contamination [Black and Talbot \(2005\)](#). However, in some developing countries, improvements in source quality have yielded minimal reductions in diarrheal disease [Bennett \(2008\)](#); [Jalan and Ravallion \(2003\)](#). This may be because improvements at the water source do not necessarily lead to significant quality gains at the point of consumption, as households may reduce private spending on in-home water treatment, offsetting the quality improvements achieved at the source [Jalan and Ravallion \(2003\)](#).

While the scientific evidence linking E. Coli to diarrhea prevalence is clear ([Nataro & Kaper, 1998](#)), there is a lack of evidence on how much of this relationship is observed in real society. Combining 21 relevant studies, the meta-analysis shows a significant reduction in diarrhea due to water treatment [Arnold, Colford Jr, et al. \(2007\)](#). The findings indicate that stored water E. Coli contamination was reduced by 80 percent (pooled relative risk: 0.20) and child diarrhea risk was reduced by 29 percent (pooled relative risk: 0.71) from the water treatment.¹ However, two recently published randomized controlled trials (RCTs) conducted in rural Bangladesh and Kenya found that chlorination tablet interventions did not significantly affect the incidence of diarrheal disease or anthropometric outcomes [Luby et al. \(2018\)](#); [Pickering, Null, et al. \(2019\)](#).

The Lancet Commission on water, sanitation, and hygiene highlights the urgent need for investments in WASH services, particularly in low- and middle-income countries, to reduce health disparities. Access to safe drinking water is critical in preventing infectious diseases, including those caused by E. Coli contamination [Commissioners of the Lancet Commission on Water, Hygiene, and Health \(2021\)](#). Another study provides updated estimates on the impact of water, sanitation, and hygiene (WASH) interventions on childhood diarrhoea in low- and middle-income

¹Read the paper to determine how they define E. Coli outcome. Is it binary?

countries (LMICs). It demonstrates that interventions such as water treatment at the point of use, improved drinking water sources, sanitation with sewer connections, and handwashing with soap can significantly reduce the risk of diarrhoea in children [Wolf et al. \(2022\)](#). These findings highlight the crucial role of WASH interventions in mitigating diarrhoeal disease, although no studies evaluated interventions achieving universal access to safely managed WASH services. The Lancet Commission on water, sanitation, hygiene, and health highlights the critical importance of improving WASH services for controlling diarrheal diseases, particularly those caused by E. Coli contamination in water. Water treatment interventions should focus not only on microbiological water quality but also address a broader spectrum of health outcomes beyond diarrhea, including mental health and gender disparities [Stoler, Guzmán, and Adams \(2023\)](#).

Recent studies suggest that passive chlorination at the point of water collection can be an effective intervention for reducing childhood diarrhoea in urban low-income settings [Pickering, Crider, et al. \(2019\)](#). Automated chlorine dosers have been shown to disinfect drinking water without requiring electricity, leading to a significant reduction in diarrhoea prevalence as defined by WHO standards. Despite these technological advancements, other low-cost water treatment interventions, such as improved cookstoves and safe water devices, often fail to achieve their expected health benefits due to inconsistent adoption by low-income households [Ray and Smith \(2021\)](#). This inconsistency hampers efforts to improve water quality, reduce E. Coli contamination, and decrease the prevalence of waterborne diseases such as diarrhoea. Additionally, the paper argues that shifting from household-scale interventions to utility-scale service models could ensure more reliable access to clean water and sanitation, which are vital in preventing diarrhoeal diseases. Overall, the global impact of inadequate WASH services is substantial, with an estimated 1.4 million deaths and 74 million disability-adjusted life years (DALYs) that could have been averted in 2019 through safe WASH practices [Wolf et al. \(2023\)](#). These findings underscore the importance of WASH interventions in addressing the global burden of diarrhoeal diseases linked to poor water quality and sanitation.

The remainder of this paper is organized as follows. Section 2 describes ...

2 Methods

2.1 Sample

To examine the relationship between source water contamination, household water treatment decisions, and contamination in stored drinking water in the same framework, we analyze nationally representative household data from the MICS conducted in 25 countries across Africa, Latin America, and Asia.² We accessed the data on September 3, 2024. Authors did not have access to any information that could identify individual participants during or after data collection. All MICS datasets are anonymized prior to public release. We use data collected between 2017 and 2021, in which a rapid water quality testing module is available. The surveys were implemented by UNICEF in collaboration with national ministries and statistics offices.

In households selected for water testing, interviewers requested permission to collect water samples from two locations: (1) the source from which the household collects water, and (2) the stored drinking water within the household. Interviewers first asked for a sample of drinking water by requesting a glass of drinking water, and then asked to be shown the water source location in order to collect a sample directly from the water source. These water samples were analyzed for the presence of *Escherichia coli* (E. Coli), an indicator of fecal contamination.³

Depending on the country, approximately 10–25 percent of a random subsample of households were selected for the rapid water quality testing module (Table A.1). Overall, 92.6 percent of households completed water testing with non-missing data, and in most countries, E. coli data are available for over 80–90 percent of the selected households. While non-consent was rare ($N = 7$), water testing was occasionally not conducted due to the household’s inability to provide a water sample, partial completion, or errors in reading the test results. A comparison of household characteristics between those who completed the water testing and those who did not shows varying patterns across countries (Appendix A). The final analytical sample includes 59,633 households from

²From an initial sample of 29 countries, we excluded Tuvalu, Kiribati, Tonga, and the Turks and Caicos Islands, where fewer than 1,000 households completed the water testing module, to ensure sufficient data for reliable country-level statistics.

³The drinking water quality module for the survey was developed collaboratively by the Joint Monitoring Program (JMP) and the global MICS program. An international water quality trainer, with support from national experts, led the training of field teams. National experts from regulatory agencies, research labs, or ministries of health provided technical input and oversight during fieldwork through a few field visits. Water samples were analyzed for E. Coli using portable membrane filtration. Field teams filtered 100mL of water through a 0.45- μ m filter, placed it on CompactDry EC growth media plates, and rehydrated it with 1mL of sample water.

25 countries.

This paper first presents the prevalence of four types of water treatment methods by country: (1) boiling, (2) use of chlorine products, Aquatabs, or PUR, (3) straining or settling, and (4) other methods. The water treatment methods analyzed refer specifically to the water treatment applied to the water sample for *E. coli* presence, which differs from the question on household water treatment practices in general.⁴

The study then presents the *E. coli* contamination levels of water at the source by country. *E. coli* levels are categorized based on the number of colony-forming units (CFUs) per 100 mL: less than 1, 1–100, and more than 100. For ease of description, this paper refers to water with less than 1 CFU as “low risk” or “free from *E. coli*,” 1–100 CFU as “moderate risk,” and more than 100 CFU as “high risk.”⁵

3 Methodology

Estimating the causal impact of household water treatment on *E. coli* contamination and child diarrhea is complicated by potential endogeneity. Households that treat their drinking water may systematically differ from those that do not along observable and unobservable dimensions, including socioeconomic status, education, access to infrastructure, and health awareness. If not properly addressed, these differences may bias conventional regression estimates.

To address these challenges, we employ **Double/Debiased Machine Learning (DDML)**, a recent causal inference framework that allows for consistent estimation of treatment effects in the presence of high-dimensional confounders (Chernozhukov et al., 2018). The DDML approach combines flexible machine learning methods for nuisance parameter estimation with orthogonalization and cross-fitting to mitigate regularization bias and overfitting. Specifically, DDML uses machine learning algorithms to estimate conditional expectation functions for the outcome and treatment variables, and then constructs orthogonal score functions to recover causal parameters of interest. In our context, these parameters capture the causal effects of household water treatment

⁴Figure ?? shows the relationship between water treatment reported for the tested water sample and general water treatment practices. Treatment rates are lower for the tested water, suggesting that households may not consistently apply the treatment methods they report using in general. While households are more likely to report using chlorine in response to the general question, boiling is more frequently reported for the water sample that was tested.

⁵World Health Organization (2017) use a more detailed classification, defining 1–10 CFU as moderate risk, 10–100 CFU as high risk, and more than 100 CFU as very high risk.

on *E. coli* contamination and diarrhea incidence. Additional implementation details are provided in Appendix C.

Formally, let Y_i denote the outcome of interest (either an indicator for *E. coli* detection in drinking water or child diarrhea), D_i represent household water treatment status, and X_i denote a vector of observed covariates. Throughout, we use Y to denote outcomes, D to denote treatment variables, and X to denote confounders.

We estimate two complementary DDML specifications: a *partially linear model* and an *interactive model*. Estimating both models allows us to assess the robustness of results to alternative assumptions regarding the functional relationship between covariates, treatment, and outcomes.

Partially Linear Model

First, we estimate a partially linear DDML model of the form:

$$Y_i = \alpha D_i + g_0(X_i) + U_i, \quad D_i = m_0(X_i) + V_i,$$

where α is the parameter of interest capturing the average causal effect of water treatment on the outcome, and $g_0(\cdot)$ and $m_0(\cdot)$ are high-dimensional nuisance functions approximated using machine learning methods. This specification allows for flexible control of confounders while maintaining a linear effect of treatment. The DDML procedure applies cross-fitting to ensure orthogonality between the estimated residuals, thereby delivering valid inference even when nuisance functions are estimated using regularized or nonparametric methods.

Interactive Model

Second, we estimate an interactive DDML model that relaxes the separability assumption between covariates and treatment:

$$Y_i = g_0(X_i, D_i) + U_i, \quad D_i = m_0(X_i) + V_i.$$

In this specification, the outcome is allowed to depend on treatment and covariates in a fully nonseparable manner. The treatment variable D_i is binary, and the parameters of interest are the Average Treatment Effect (ATE).

This interaction model captures potential effect heterogeneity arising from interactions between

household characteristics and water treatment behavior. Given that the effects of water treatment may vary across rural and urban settings as well as across country-specific contexts, this specification provides greater flexibility for identifying the causal impact of water treatment on the outcome variables.

By estimating both the partially linear and interactive specifications, we provide a comprehensive assessment of the causal impact of household water treatment under alternative structural assumptions. For a more detailed discussion of the methodology and implementation, see our working paper.

The study employ LASSO Random Forest and best fit model measured with that give the smallest MSPEs. If your orthogonal estimator is correctly implemented: $\hat{\theta}_t$ should not change much across ML methods.

Control variables

The **Multiple Indicator Cluster Surveys (MICS)** provide an ideal setting for applying DDML. With more than 59,000 households from 25 countries, the data offer substantial variation in both exposure to water treatment and household characteristics. The surveys also contain a rich set of potential confounders—covering demographic, socioeconomic, environmental, and health behavior dimensions—that can be flexibly incorporated into the high-dimensional control set. This richness allows the DDML framework to effectively mitigate selection bias while leveraging the large sample size to improve estimation precision.

In the empirical implementation, we estimate the treatment effect of household water treatment separately for (i) *E. coli* contamination for the household drinking water and (ii) reported child diarrhea incidence, using cross-fitted DDML estimators with lasso-based nuisance models.

More specifically, the treatment variable captures household water treatment practices, including boiling and chlorination. The set of control variables includes household wealth indices constructed separately for urban and rural areas, an urban–rural indicator, the education level of the household head, *E. coli* counts at the water source, water source type, and country fixed effects.⁶

Analysis 2

- Y: Diarrhea among children

⁶Additional controls not reported include age, sanitation facilities, household housing quality, and other household characteristics.

- T: E. coli for drinking water (Water treatment)
- We have E. coli at the water source
- We have arsenic levels of water (where literature says it is highly correlated with E.coli contamination), but water treatment does not change the arsenic level. Maybe good control for the quality of the water (not affected by the water treatment)
- Various other controls: Age, Education, country, toilet, household quality etc

4 Descriptive Statistics

Table 1 presents household characteristics and water treatment behavior based on the contamination level of the source water. The final sample size is 59,633 households from 25 countries, with 25,518 (43 percent) in the low-risk category, 21,844 (37 percent) in the moderate-risk category, and 12,271 (21 percent) in the very high-risk category at the water source.

4.1 E. Coli in drinking water and diarrhea

Figure 1 shows the relationship between E. Coli counts in source and drinking water and the incidence of diarrhea and fever among households. The histograms show the distribution of E. Coli counts in both source and drinking water. The green histogram indicates that the E. Coli count in source water peaks at lower values, while the black histogram shows a similar distribution for drinking water. This distribution indicates that drinking water is more likely to be contaminated than source water, suggesting that water becomes contaminated during storage.

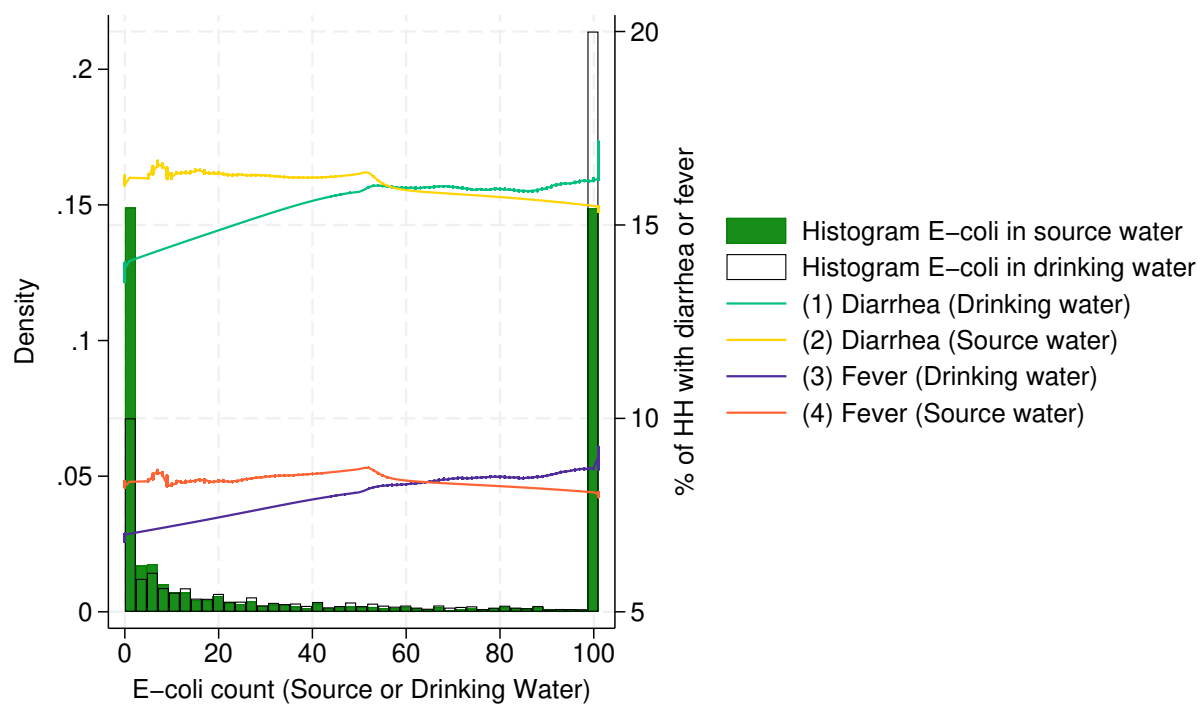
The figure shows that the diarrhea rate is highly correlated with drinking water quality. The average probability of children under age 5 having diarrhea is around 11 percent when the water is not contaminated (E. Coli = 0), but the rate increases to as high as 16 percent if the water is contaminated at levels higher than E. Coli over 100. The contamination level of the water source is unrelated to the prevalence of diarrhea, indicating that households are treating water to reduce E. Coli based on different initial contamination levels. A similar relationship is observed for the incidence of fever.

Table 1: Household Characteristics by Water Source Contamination Level (share)

	Low risk	Moderate risk	High risk
Urban	0.53	0.31	0.18
Socioeconomic level			
Poorest	0.14	0.24	0.31
Poor	0.16	0.23	0.25
Middle	0.18	0.21	0.21
Rich	0.23	0.18	0.15
Richest	0.28	0.14	0.07
Primary water source			
Piped water	0.54	0.27	0.10
Tube/Well/Borehole	0.24	0.28	0.11
Protected well/spring	0.06	0.11	0.13
Unprotected well/spring	0.06	0.23	0.44
Surface/Rain water	0.01	0.09	0.21
Packaged/Bottled water	0.07	0.02	0.01
Others	0.01	0.01	0.01
Any treatment (Water tested)	0.13	0.13	0.17
Nothing	0.87	0.87	0.83
Boil	0.02	0.02	0.02
Chlorine/Aquatabs/PUR	0.02	0.02	0.02
Strain/Settle	0.09	0.09	0.11
Other	0.01	0	0.01
Water test results			
Source water test (100ml)	0	23.02	101
Household water test (100ml)	23.88	49.43	90.66
N	6,569	8,450	7,391

Notes: The table presents the household characteristics across 25 countries. The number of CFUs per 100mL of E. Coli is capped at 101 if the number is higher than 100. The mean of the blank water test is 0.67 CFUs per 100mL.

Figure 1: Relationship between E. Coli of Source/Drinking Water and Diarrhea and Fever



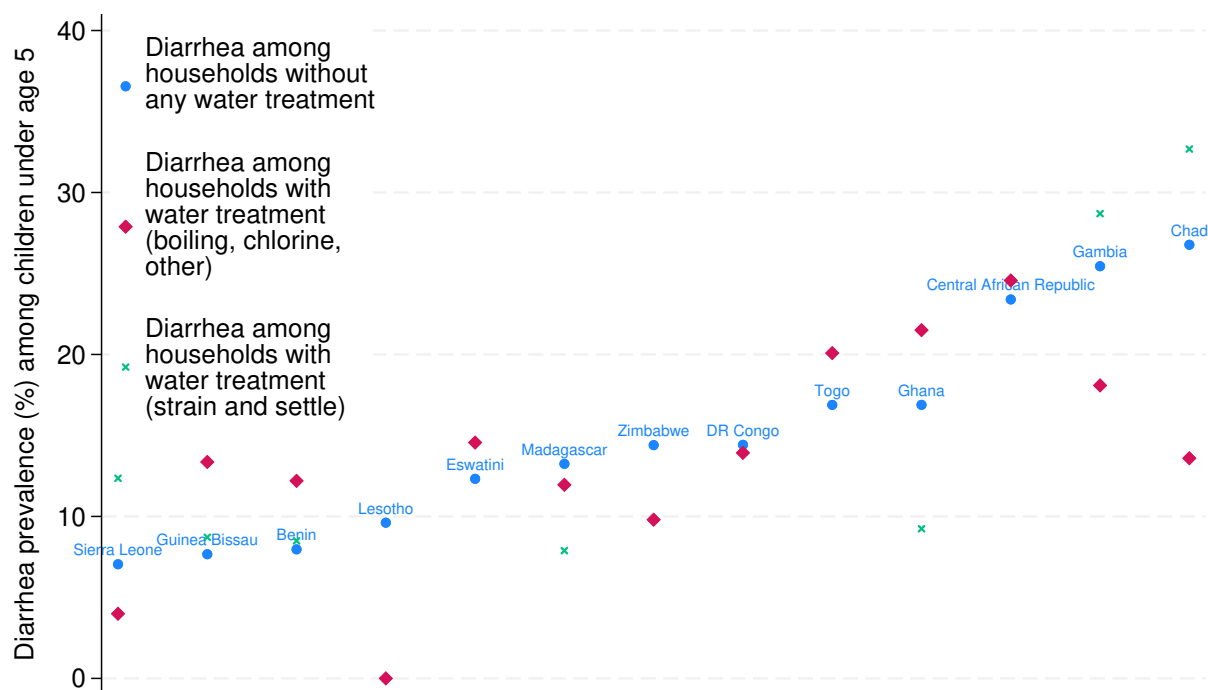
Notes: N=22,589 children under age five. The figure shows the relationship between E. Coli counts in source and drinking water and the incidence of diarrhea and fever among households. The histograms show the distribution of E. Coli counts in both source and drinking water.

4.2 Water treatment and diarrhea

Figure 2 shows the prevalence of diarrhea among children under age 5, grouped by household water treatment methods. While some countries exhibit a lower prevalence of diarrhea among households that treat their water, the difference is minor. In some instances, the relationship is reversed, with a higher prevalence of diarrhea observed among households that treat their water. Such patterns are widely observed, and several studies using MICS data do not find a clear link between water treatment and diarrhea (Husein, Nounkeu, Armah, & Dharod, 2023).

The unclear relationship can be attributed to endogeneity. Households with contaminated water, which is correlated with higher diarrhea prevalence, are more likely to treat their water, while those using uncontaminated water sources are less likely to do so. Another reason for the weak link is the low efficacy of the water treatment itself.

Figure 2: Diarrhea prevalence among U5 children by water treatment methods



Notes: The prevalence of diarrhea among children under age 5 across 14 countries from MICS, using nationally representative weights. Categories with fewer than 15 samples are excluded from the analysis, as the sample size is too small to calculate a reliable average.

5 Empirical Results

This section presents the ordinary least squares (OLS) estimates as a benchmark, followed by results from the Double/Debiased Machine Learning (DDML) framework using both the partially linear and interactive specifications.

5.1 OLS

Table 2 presents the ordinary least squares (OLS) estimates using the same set of control variables as in the machine learning specifications. The results indicate that boiling reduces *E. coli* counts by approximately 12.8–13.6 units and lowers the probability of high-risk drinking water by 13.0–13.3 percentage points. Similar reductions are observed for other water treatment methods, although statistical significance is primarily found when *E. coli* is measured as a continuous outcome.

When water treatment methods are grouped into boiling and chlorination, the estimates show a reduction in *E. coli* counts of approximately 12.0 units and a decrease in the likelihood of high-risk drinking water (defined as *E. coli* >100) by 11.5 percentage points.

Table 2: E-coli in drinking water

	E-coli (high risk)		E-coli (some risk)		Diarrhea	
	(1)	(2)	(3)	(4)	(5)	(6)
Any water treatment		-0.137*** (0.017)		-0.045*** (0.006)		-0.020 (0.015)
Boiling	-0.134*** (0.027)		-0.057*** (0.009)		-0.010 (0.024)	
Chlorination	-0.167*** (0.025)		-0.036*** (0.008)		-0.032 (0.022)	
Strain	-0.032* (0.017)		-0.012** (0.006)		0.024* (0.015)	
Other treatment	-0.074 (0.047)		-0.053*** (0.016)		0.007 (0.042)	
Do not know	0.306 (0.233)		0.017 (0.081)		0.227 (0.206)	
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Mean	0.60	0.60	0.98	0.98	0.16	0.16
Adjusted R^2	0.32	0.32	0.05	0.05	0.04	0.04
Observation	14,141	14,130	14,141	14,130	13,940	13,929

Notes: Observations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 DDML

Table 3: Diarrhea: OLS vs. DDML (LASSO / RF) and BEST — Partial Effect of D on Y

	(1) OLS
Any treatment (Water tested)	-0.060*** (0.012)
Observations	19,562
Reps	5
D_water_treatment_ss_mse	
Y_SomeRiskHome_ss_mse	

Table 4: Diarrhea: OLS vs. DDML (LASSO / RF) and BEST — Partial Effect of D on Y

	(1) OLS
Any treatment (Water tested)	-0.084*** (0.016)
Observations	19,562
Reps	5
D_water_treatment_ss_mse	
Y_SomeRiskHome_ss1_mse	
Y_SomeRiskHome_ss0_mse	

6 Conclusion

Contributors

Declaration of interests

All authors declare no competing interests.

Data sharing

All data used for this and analytical codes are available immediately following publication without an end date to anyone for any purpose and are either published in the appendices or can be accessed through the corresponding author.

Table 5: Diarrhea: OLS vs. DDML (LASSO / RF) and BEST — Partial Effect of D on Y

	(1) OLS
Any treatment (Water tested)	-0.133*** (0.016)
Observations	19,562
Reps	md
D_water_treatment_ss_mse	
Y_WQ26_ss_mse	

Table 6: Diarrhea: OLS vs. DDML (LASSO / RF) and BEST — Partial Effect of D on Y

	(1) OLS
Any treatment (Water tested)	-0.133*** (0.022)
Observations	19,562
Reps	md
D_water_treatment_ss_mse	
Y_WQ26_ss1_mse	
Y_WQ26_ss0_mse	

Table 7: Diarrhea: OLS vs. DDML (LASSO / RF) and BEST — Partial Effect of D on Y

	(1) OLS
Any treatment (Water tested)	-0.002 (0.014)
Observations	19,562
Reps	5
D_water_treatment_ss_mse	
Y_diarrhea_ss_mse	

Table 8: Diarrhea: OLS vs. DDML (LASSO / RF) and BEST — Partial Effect of D on Y

	(1) OLS
Any treatment (Water tested)	-0.013 (0.017)
Observations	19,562
Reps	5
D_water_treatment_ss_mse	
Y_diarrhea_ss1_mse	
Y_diarrhea_ss0_mse	

Acknowledgments

References

- Arnold, B. F., Colford Jr, J. M., et al. (2007). Treating water with chlorine at point-of-use to improve water quality and reduce child diarrhea in developing countries: a systematic review and meta-analysis. *American Journal of Tropical Medicine and Hygiene*, 76(2), 354–364. <https://doi.org/10.4269/ajtmh.2007.76.354>
- Bennett, D. (2008). Does clean water make you dirty? Water supply and sanitation in the rural Philippines. *Journal of Human Resources*, 43(4), 694–721. <https://doi.org/10.1353/jhr.2012.0008>
- Black, M., & Talbot, D. (2005). *Water, sanitation, and hygiene for all: Partnerships and innovations for the next decade*. UNICEF.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). *Double/debiased machine learning for treatment and structural parameters*. Oxford University Press Oxford, UK.
- Commissioners of the Lancet Commission on Water, S., Hygiene, & Health. (2021). The Lancet Commission on water, sanitation, and hygiene, and health. *The Lancet*, 398(10304), 1315–1348. DOI:10.1016/S0140-6736(21)02005-5
- Husein, M., Nounkeu, C. D., Armah, S., & Dharod, J. M. (2023). Examination of water quality at the household level and its association with diarrhea among young children in Ghana: Results from UNICEF-MICS6 survey. *PLOS Water*, 2(6), e0000049. <https://doi.org/10.1371/journal.pwat.0000049>
- Jalan, J., & Ravallion, M. (2003). Does piped water reduce diarrhea for children in rural India? *Journal of Econometrics*, 112(1), 153–173. [https://doi.org/10.1016/S0304-4076\(02\)00158-6](https://doi.org/10.1016/S0304-4076(02)00158-6)
- Luby, S. P., Rahman, M., Arnold, B. F., Unicomb, L., Ashraf, S., Winch, P. J., ... others (2018). Effects of water quality, sanitation, handwashing, and nutritional interventions on diarrhoea and child growth in rural Bangladesh: a cluster randomised controlled trial. *The Lancet Global Health*, 6(3), e302–e315. [https://doi.org/10.1016/S2214-109X\(18\)30031-8](https://doi.org/10.1016/S2214-109X(18)30031-8)
- Nataro, J. P., & Kaper, J. B. (1998). Diarrheagenic escherichia coli. *Clinical Microbiology Reviews*, 11(1), 142–201. <https://doi.org/10.1128/CMR.11.1.142>

- Pickering, A. J., Crider, Y., Sultana, S., Swarthout, J., Goddard, F. G., Islam, S. A., ... Luby, S. P. (2019). Effect of in-line drinking water chlorination at the point of collection on child diarrhoea in urban Bangladesh: a double-blind, cluster-randomised controlled trial. *The Lancet Global Health*, 7(9), e1247–e1256. [https://doi.org/10.1016/S2214-109X\(19\)30315-8](https://doi.org/10.1016/S2214-109X(19)30315-8)
- Pickering, A. J., Null, C., Winch, P. J., Mangwadu, G., Arnold, B. F., Prendergast, A. J., ... others (2019). The WASH Benefits and SHINE trials: interpretation of WASH intervention effects on linear growth and diarrhoea. *The Lancet Global Health*, 7(8), e1139–e1146. [https://doi.org/10.1016/S2214-109X\(19\)30268-2](https://doi.org/10.1016/S2214-109X(19)30268-2)
- Ray, I., & Smith, K. R. (2021). Towards safe drinking water and clean cooking for all. *The Lancet Global Health*, 9(3), e361–e365. doi: 10.1016/S2214-109X(20)30476-9
- Stoler, J., Guzmán, D. B., & Adams, E. A. (2023). Revisiting transformative WASH: measuring impact. *The Lancet Global Health*, 11(4), e493–e494. [https://doi.org/10.1016/S2214-109X\(23\)00013-X](https://doi.org/10.1016/S2214-109X(23)00013-X)
- Wolf, J., Hubbard, S., Brauer, M., Ambelu, A., Arnold, B. F., Bain, R., ... others (2022). Effectiveness of interventions to improve drinking water, sanitation, and handwashing with soap on risk of diarrhoeal disease in children in low-income and middle-income settings: a systematic review and meta-analysis. *The Lancet*, 400(10345), 48–59. [https://doi.org/10.1016/S0140-6736\(22\)00937-0](https://doi.org/10.1016/S0140-6736(22)00937-0)
- Wolf, J., Johnston, R. B., Ambelu, A., Arnold, B. F., Bain, R., Brauer, M., ... Cumming, O. (2023). Burden of disease attributable to unsafe drinking water, sanitation, and hygiene in domestic settings: a global analysis for selected adverse health outcomes. *The Lancet*, 401(10385), 1861–1873. [https://doi.org/10.1016/S0140-6736\(23\)00458-0](https://doi.org/10.1016/S0140-6736(23)00458-0)
- World Health Organization. (2017). *Guidelines for Drinking-Water Quality, Fourth Edition, Incorporating the First Addendum*. Author. <https://apps.who.int/iris/rest/bitstreams/1080656/retrieve> (Retrieved June 1, 2024)

Appendices

A Sample

Table A.1 reports the total number of households in the full survey, the number selected for the water quality testing module, and the number that provided consent. It also shows the number of samples with missing values for either water source or stored drinking water quality data, the number of samples missing water treatment information, and the final number of samples included in the analysis.

B Continuous

Table A.1: Number of Household Surveys and Sample Included in the Analysis

	Toatl	Selected		Not in the sample			Complete	
		Number	Share (%)	No consent	Water test missing	Water treat-ment missing	Number	Share (%)
Sub-Saharan Africa								
Central African Rep	8133	1228	15.1	1	190	1	1036	84.4
Chad	18967	2284	12	0	171	0	2113	92.5
Zimbabwe	11091	2138	19.3	0	144	2	1992	93.2
Gambia	7405	1879	25.4	0	127	1	1751	93.2
Togo	7916	1157	14.6	0	73	2	1082	93.5
Eswatini	4675	1235	26.4	1	70	4	1160	93.9
Madagascar	17870	3439	19.2	0	175	4	3260	94.8
Benin	16790	3809	22.7	0	152	8	3649	95.8
Lesotho	8847	1376	15.6	0	44	0	1332	96.8
Ghana	12939	3230	25	0	96	3	3131	96.9
Malawi	25419	3208	12.6	0	87	5	3116	97.1
DR Congo	20792	2783	13.4	0	77	0	2706	97.2
Sierra Leone	15309	1784	11.7	0	37	4	1743	97.7
Guinea Bissau	7379	1829	24.8	0	6	1	1822	99.6
Latin America								
Honduras	20669	5111	24.7	0	1088	2	4021	78.7
Dominican Rep	31488	3125	9.9	0	581	3	2541	81.3
Suriname	7915	1982	25	2	362	2	1616	81.5
Guyana	7072	1670	23.6	0	271	6	1393	83.4
Trinidad & Tobago	7499	1905	25.4	0	301	3	1601	84
Fiji	5467	1104	20.2	0	18	0	1086	98.4
Asia								
Mongolia	13798	2764	20	0	188	4	2572	93.1
Nepal	12655	2547	20.1	1	153	12	2381	93.5
Lao	22287	3360	15.1	0	112	0	3248	96.7
Viet Nam	13359	3323	24.9	0	90	2	3231	97.2
Bangladesh	61242	6150	10	2	98	2	6050	98.4
Total	386983	64420	16.6	7	4711	71	59633	92.6

Notes: Households that did not consent to the full survey are excluded from the sample, as household characteristics are unavailable for analysis. Cases with missing water test data reflect instances where water testing was not conducted—either because the household could not provide a water sample, the test was only partially completed, or there were errors in reading the results.

Table B.1: Diarrhea: OLS vs. DDML (LASSO / RF) and BEST — Partial Effect of D on Y

	(1) OLS
Any treatment (Water tested)	-12.876*** (1.429)
Observations	19,562
Reps	md
D_water_treatment_ss_mse	
Y_WQ26_ss_mse	

Table B.2: Diarrhea: OLS vs. DDML (LASSO / RF) and BEST — Partial Effect of D on Y

	(1) OLS	(2) LASSO
Any treatment (Water tested)	-14.203*** (1.993)	-14.203*** (1.993)
Observations	19,562	19,562
Reps	md	md
D_water_treatment_ss_mse		
Y_WQ26_ss1_mse		
Y_WQ26_ss0_mse		

C DDML Command

The DDML framework is particularly useful when estimating causal effects in high-dimensional settings, as it allows for flexible machine learning control while preserving valid inference through sample-splitting and orthogonalization.

This section provides guidance on how to use the `qddml` command, which implements the Double/Debiased Machine Learning (DDML) framework proposed by [Chernozhukov et al. \(2018\)](#). The official documentation can be found at the following link: [qddml Help File](#).

Step 1: Set up Python integration in Stata. Before running any DDML commands, you must configure Stata’s Python integration. Follow the official setup instructions available here: [Python Configuration for Stata](#). This step ensures that DDML can communicate properly with Python-based learners.

Step 2: Run the analysis do-file. Once Python integration is set up, you can execute the analysis script:

[Good YouTube video](#)